International Journal of Data Science and Analytics (2026) 22:227
https://doi.org/10.1007/s41060-026-01202-w

REGULAR PAPER

An optimal CSBO-based detection and classification of breast cancer using deep SAC-LSTM design from histopathological image

T. Grace Shalini[1] · **S. Rathnamala**[2]

Received: 11 December 2025 / Accepted: 17 June 2026
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Abstract
Cancer is regarded as one such top mortality cases that occurs with the abnormal growth of cells in human body. It becomes a great challenge causing huge threat to the people's safety and their wellbeing's throughout the world. For analyzing an image, the study of biopsy (histopathology) is carried most often that aids in the detection of breast cancer. The main intention of this work is to present an improved model for different phases of Computer Aided Diagnosis (CAD) system which plays a significant and challenging role on minimizing the gap between the existing research. An average weighted-based Gaussian filtering approach is employed for the purpose of pre-processing which is followed by Cauchy distribution-based segmentation of pre-processed image. From the segmented image, the features are extracted with the use of an optimal feature extraction model termed conjugate sword bee optimization-based feature extraction to attain optimal features. In addition, an improved classification design to detect breast cancer is employed as self-attention mechanism coupled with Long Short Term Memory classifier approach that is SAC-LSTM to diagnose and classify the kind of breast cancer like malignant or benign. In practice, the application of presented optimal feature extraction and SAC-LSTM classifier model for CAD system thus reduces the number of incorrect detections and diagnoses thereby increasing the classifier accuracy. The BreakHis dataset consisting of histopathological image is considered as an input for the performance estimation. Hence, this framework might serve as an effective opinion strategy for the pathologists thus aiding in the diagnosis of disease at an early stage.

Keywords Cancer · Breast cancer detection · Histopathology images · Average weighted Gaussian filtering · Segmentation · CSBO-based feature extraction · SAC-LSTM · CAD system · Classification accuracy

1 Introduction

In today's world, Breast cancer is considered as a life-threatening disease that has an adverse effect on lives of many women. This is also considered as a second important reason for deaths that occurs among females due to cancer [1]. As of report Breast cancer seems to affect 1 million individuals annually which results in about 400,000 mortality all over the globe. It is either a benign or malignant tumorous one that is instigated by uncontrolled division and growth of cells within the breast [2–5]. The tumors that are non-cancerous are regarded as benign those that are not a life-threatening ones which are treatable with medication. The tumors that are cancerous are regarded as a malignant one, and if it is left untreated, the patient might die. Those growths occur as a lump in breast and it could be identified with the use of X-rays. So as to detect and recognize them at earlier stage, patients should consult their healthcare provider immediately in case the lump appears in their breast. The following factors could be the reason and are a leading cause for the occurrence of breast cancer: their family history, genes, age, lifestyle and dietary habits. The cost of treatment is thus higher for the breast cancer which causes unadorned side effects afterward long-term medications use by the patients [6, 7]. Several models occur because of detection of cancer at early stage where it is not possible for curing ailment. Early disease detection is thus crucial according to the health professionals. Therefore, the pathologists might

✉ T. Grace Shalini
gracesht@srmist.edu.in

S. Rathnamala
rathnamala@sethu.ac.in

[1] Department of Computational Intelligence, School of Computing, S.R.M. Institute of Science and Technology, Kattankulathur, Chennai, T.N., India

[2] Department of Artificial Intelligence and Data Science, Sethu Institute of Technology, Kariapatti, T.N., India

Published online: 09 July 2026

face a surge in work considerably for the diagnosis of the histopathological image of cancer. Recently, the development of CAD systems aids in minimization of workload. The digital pathology remains to rise as a wide-reaching energy for the purpose of diagnosis. The technique of DL (deep learning) was developed nowadays for solving huge problems in the medical image processing area [8]. Also, in these days, healthcare practitioners too have developed a huge attention in the detection and prediction of breast cancer by means of machine learning (ML) approaches. On the contrary, another model named DL which is a subcategory of ML technique, at which the approach might make decision on its own whether the prediction is precise or not. The deep learning-based model is employed in this work so as to diagnose the breast cancer type (benign or malignant). A small tissue portion is taken from the area suspected during biopsy which undergoes lab test to confirm the cancer cell presence. Based on the location, size, and cancer number present, type of biopsy will be selected.\query{Please check the edit made in the article title.}

The subsequent portion of presented manuscript is prearranged as shown: Section 2 is a review analysis made on various existing methods related to breast cancer detection. Section 3 is the brief narration on the proposed method implemented. Section 4 offers a presentation of performance analysis estimated. Section 5 concludes the overall work implemented.

2 Related works

In the work [9], a novel technique for the breast cancer detection was employed using ML technique. An experimental study was carried over the input dataset for evaluating the performance. The suggested method offers effective outcomes with high accuracy on comparing traditional methods.

A DL aided effective Adaboost algorithm (DLA-EABA) was presented in [10] for detection of breast cancer which was presented mathematically with the advanced computational methods. Along with the traditional approach of computer vision, methods of classifying tumor using transfers were actively developed by deep CNNs (convolutional neural networks). The approach starts by evaluating CNN-dependent transfer learning for characterizing tissues of breast intended for various predictive, diagnostic or prognostic tasks otherwise various imaging modalities like MRI, mammography, US (Ultrasound), and DBT (digital breast tomosynthesis). An experimental outcome reveals that the proposed method offers higher rate of accuracy (97.2%), 98.3% sensitivity, and 96.5% specificity on comparing traditional methods.

An annotation-effective deep learning approach was presented in [11] which achieves existing mammogram classification performance, encompasses to DBT; 3D mammography successfully, and thereby detects the tumors, well simplifies for a population having lower screening rates, and finally outdoes well for full-time breast imaging specialists on improving the sensitivity absolutely on an average of 14%. The outcomes reveal a promise over software which enhances the accuracy and admittance to the showing mammography all over the world.

In [12], the author presented a review on ML and DL approaches together with the desired breast cancer application. An existing approach of ML shows the use of limited techniques, while the approaches of DL were having huge latent intended for the application in a medical study for enhancing the diagnostic capability of existing CAD system. By the review, it was identified that a varied breast density masses was a more challenging one for detection and classification on comparing with calcification. A traditional ML technique offers confined approaches that were limited to neither specific type of density nor datasets. Though, the DL technique shows a promising enhancement in the analysis of breast cancer, there are some issues like computational cost and data scarcity that has to be overwhelmed to a great level on employing augmentation of data and enhanced computation power of the DL approaches.

A CAD system depending on the mammogram was presented in [13] which enables primary diagnosis, detection and breast cancer treatment. Moreover, a traditional CAD system's accuracy still remains an unsatisfactory one. A method for the detection of breast cancer based on the method of feature fusion was explored in this work using deep CNN features. Initially, a method of mass detection depending on CNN deep features and the ELM that are unsupervised (extreme learning machine) clustering approach was presented. Then, a feature set to fuse deep feature, texture feature, density feature, morphology feature was build. At last, classifier ELM was presented with the use of feature set that were fused for classifying malignant and benign breast masses. An experimental analysis to demonstrate the efficiency and accuracy of suggested mass diagnosis and classification of breast cancer was introduced.

A review on multi-layer perceptron (MLP), CNN, and Artificial neural network (ANN) was employed in [14] for the detection of breast malignancies to diagnosis breast cancer earlier depending on their accuracy so as to recognize which technique was better in diagnosing malignancies of breast cell. Deep comparison on functioning of each network with their scheming was accomplished, and after that the analysis was made depending on the diagnostic accuracy and breast malignancy classification by network for deciding which network performs well than others. From the analysis, CNN was identified to offer a high rate of accuracy on

comparing MLP diagnosis and the breast cancer detection. Still, there was a necessity of analyzing and performing some careful research that employs both techniques on a same dataset at some conditions so as to identify architecture which offers better accuracy.

A review was made in [15] for exploring the ways at which the DL was better integrated into the screening workflows of breast cancer with the use of DBT. Initially, the principles behind DBT were explained and for what reason this becomes a gold standard in screening breast. Then, a review on foundation of the DL method in the image diagnostic, assessment on present state-of-research on AI dependent DBT interpretation was carried. Lastly, some limitations on the integration pf some AI techniques to clinical practice and an opportunity of these were presented in the field of burgeoning.

In [16], a new technique of CAD system was proposed to classify malignant and benign tumors at the images of breast mammography. By the system of CAD, 2 approaches of segmentation were employed. First technique includes region of interest (ROI) manual determination, whereas the subsequent approach employs a region- and threshold-based technique. A deep CNN was employed for the extraction of features. A well-renowned architecture of DCNN termed AlexNet was employed and is fine-tuned for the classification of two classes in spite of thousand classes. A final fc layer (fully connected) is thus linked to the classifier of SVM (Support Vector Machine) so as to attain enhanced accuracy rate.

A novel approach termed BCD-WERT was suggested in [17] which uses ERT (Extremely Randomized Tree) and a WOA (Whale Optimization Algorithm) intended for the selection of feature and classification process. This WOA decreases dimensionality of dataset and thus excerpts appropriate features for the precise classification. An experimental evaluation on traditional comprehensive dataset demonstrates enhanced performance on comparing them with 8 other ML techniques. From the analysis, it was evident that BCD-WERT method performs better by offering high rate of accuracy (99.30%) which was followed by SVM which offers accuracy rate of 98.60%.

In [18], the work aims at comparing the detection of breast cancer performance of the radiotherapist's interpretation on the mammographic investigations that are unassisted vs the one that are maintained by AI system. The digital mammographic examinations screening from about 240 women of median age group, 62 years, and at 39–89 years range were performed among the year 2013 and 2017. From 240 examinations, the following interpretations were got 100 showing cancers, 100 normal, and 40 leading to recalls of false positive. This interpretation was made by 14 Mammogram Quality Standards Act with Skilled radiotherapists with and without the support of AI system. The performance was estimated with respect to AUC (area under curve), sensitivity, specificity, and reading time and the analysis was compared among conditions on employing mixed-models examination of the modification and linear comprehensive representations for several repeating extents.

A comprehensive model was presented in [19] for locating the region of in the image of mammogram. The presented technique provides reduction of noisy image, segmentation of optimal image depending on CNN, optimized feature extraction, grasshopper optimization algorithm (GOA), and feature selection by GOA, thus enhancing precision and decreasing the cost of computation. The final outcomes shows that the presented model attains Sensitivity of 96%, Specificity of 93%, PPV of 85%, NPV 97%, and accuracy rate of 92%, with better efficiency on comparing existing approaches with respect to specificity, sensitivity, NPV, accuracy and PPV.

A novel framework based on deep learning was suggested in [20] to detect and classify breast cancer in the images of cytology of breast using a transfer learning concept. Generally, the architecture of DL was thus exhibited as a desired problem and was accomplished in the inaccessibility. The images feature was mined with the use of pre-trained designs of CNN in the proposed model termed GoogLeNet, VGG-Net, ResNet that were fed into the fully connected layer for the benign and malignant cells classification with the use of average pooling classification. The outcomes reveal that the suggested framework outperforms other DL architectures with respect to accurateness in the classification and detection of breast tumors in the images of cytology.

Several models occur because of detection of cancer at early stage where it is not possible for curing ailment. Early disease detection is thus crucial according to the health professionals. Therefore, the pathologists might face a surge in work considerably for the diagnosis of the histopathological image cancer. Hence, there is a need to implement effective models.

3 Proposed work

The implemented methodology is narrated in detail in this section. The entire workflow of suggested methodology is given below (Fig. 1):

3.1 Average weighted Gaussian filtering for pre-processing

Primarily, the input dataset was pre-processed so as to eliminate surplus noise existing in an image. The noise's presence might disturb the entire image quality. Therefore, for enhancing image quality and for getting enhanced outcome which results in furthermore image processing for which the

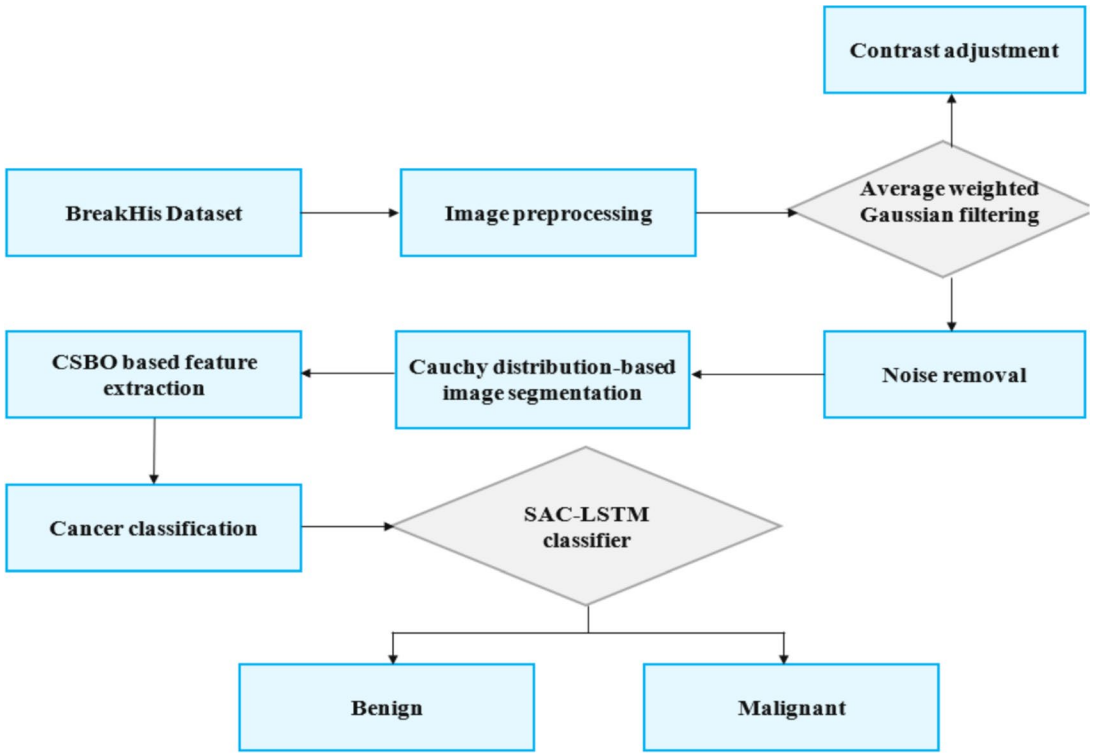


Fig. 1 Proposed methodology work flow

pre-processing is carried. After that, the images scanned are thus pre-processed using filtering approach. In this, Gaussian filtering is employed mainly for smoothening the textures, boundaries conservation, and thus Gaussian noise reduction. The average of weighted intensity from neighborhood pixel was thus located at each image pixel. This weight thus depends on Euclidean distance and too on the radiometric variation like color intensity and depth distance.

The relation of filter is thus given as follows:

$$AF[I]_p = \frac{1}{w_p}\sum_{p,q\in s} G_{\sigma s}(\|p-q\|)G_{\sigma r}(\|I_p - I_q\|) \tag{1}$$

$$G_\sigma(y) = \frac{1}{\sigma\sqrt{2\pi}}\exp\left(\frac{-y^2}{2\sigma 2}\right) \tag{2}$$

Here, Wp refers to the normalization factor, G σs denotes the weighted average that are normalized, I signify image, $\sigma sand\sigma v$ denotes the parameters of filtering. Such a reduction in noise is considered as a distinctive step of pre-processing for enhanced performance. Typically, pre-processing enhances the contrast of image. Histogram equalization (HE) is carried for improving the quality of image and it is a computerized method meant for enhancing image contrast. This is attained by growing most normal values of intensity effectively. The distinction among the extents is thus condensed few locals in difference. Therefore, an image contrast in an average is thus augmented on following technique of histogram equalization which is a procedure which needs intensity adjustment, thereby enhancing contrast which is specified by Q as an image uniform histogram for each possible intensity. It is notated as shown:

$$\mathrm{HE(Q)} = \frac{\mathrm{Pixel}^{'}\mathrm{sIntensitysum}}{\mathrm{Averagenumberofpixel}} \tag{3}$$

This technique is thus employed to enhance the quality of image with the HE. After that, on changing values of RGB, sensitivity might be calibrated.

3.2 Segmentation based on Cauchy distribution approach

An approach of Cauchy distribution is employed for the process of segmentation. Segmentation refers to the process that partitions digital images to multiple segments that are set of pixels, also termed as image objects. Typically, the segmentation of image is employed in locating the boundaries and objects (curves, lines and so on). This section offers transitory delivery on the overview of Cauchy distribution scheme. This scheme is considered as a distinct context of family SαS. Not like the distribution of SαS that lacks in the expression of condensed logical probability distribution

function (PDF), a univariate Cauchy distribution having PDF is defined as shown:

$$p_\gamma(y) = \frac{1}{\pi}\left[\frac{\gamma}{(y-\delta)^2+\gamma^2}\right] \tag{4}$$

Here, δ refers to the location parameter that specifies location of peak distribution and γ signifies the dispersion distribution which regulates the spread of distribution over its location parameter. In the region merging condition, features of region were thus evaluated with the use of Cauchy density symmetric function that are having zero mean ($\delta=0$). The histogram coefficient details at desired sub-band is thus form fitted using Cauchy model that are assessed at assessment by using GGD (Generalized Gaussian Distribution). Through graph investigation, sub-band coefficient distribution thus decays slowly on the tails altogether. The Cauchy model thus offers an enhanced fit on those considerable ends when relating Gaussian archetypal comprehensively. The bivariate isotropic Cauchy distribution is regarded as an subordinate of multi-dimensional isotropic stable family of distribution where the PDF is expressed as follows:

$$p_\gamma(y_1, y_2) = \left[\frac{\gamma}{2\pi(y_1^2+y_2^2+\gamma^2)^{3/2}}\right] \tag{5}$$

Here, the location parameters are predictable as a zero through the additional progresses are at the framework of wavelet examination. This technique is utilized at the segmentation stages which offers high outcome rates by means of taking not only the heavy tailed behavior distribution of sub-band; however, the statistically robust dependencies over wavelet coefficient at the dissimilar scales. The feature extraction is carried after this segmentation process which also integrates an optimization model to excerpt the features that are optimum from a segmented image.

3.3 Conjugate Sword Bee optimization-based feature extraction

The process of feature extraction is carried subsequent to the process of segmentation. An essential features to detect the affected region of breast is extracted by the feature extraction technique. The optimal range of features is extracted with the help of conjugate sword bee optimization approach (CABC). A conjugate sword bee optimization scheme thus replicates the bee colonies foraging behavior from which the collective intelligence for the nectar quantity optimization in the hives arise. There are three sorts of functional grouping in the unaffected bee colony like scout bees, employed bees, and spectator bees.

The employed bees thus make a half way through the colony, whereas observers will make up another half. The employed bee thus visits nectar supply that are known previously always just before choosing a newer source of food depending on the visual information from their surroundings. In case the nectar amount in new location will be greater than preceding one's amount, the original site is memorized by bee and is not capable of remaining old ones. The onlooker bees are waiting at hives in search of hired bee that went to search best location of food source currently which is exploited. Once the process of search is stagnant, then the employed bees turn out to be a scout that begins to look for new job.

The conjugate sword bee optimization approach thus creates a dispersed initial population of the SN solutions randomly at the initial phase. As per the equation provided below, each solution is thus created with their limitations.

$$y^j_{i=}y^j_{\min} + rand[0,1]\left(y^j_{\max} - y^j_{\min}\right) i = 1, 2, \ldots, SN$$
$$J = 1, 2, \ldots D \tag{6}$$

Here, $y^j_{\min}$ and $y^j_{\max}$ signify maximum and minimum parameters D and J, respectively, which is regarded as an optimization parameter number. After initialization, population solution is thus subjected to repeated cycles $C = 1, 2, \ldots, MCN$ of the search process of the employed bees, onlooker bees, and scout bees. Each and every bee is responsible for producing values of new solution *vij* as like equation that evaluates the fitness function values.

$$Vij = yij + \phi ij(yij - ykj) \tag{7}$$

In this, $k \in \{1, 2, ..., SN\}$ and $j \in \{1, 2, ..., D\}$ represent the two indices that are chosen randomly. Although, k must be verified from$i$. ϕij is regarded as a random number among$[-1, 1]$. As soon as the information shared by employed bees, the corresponding onlooker thus recognizes new solution v_{ij} at neighborhood *yi* with the use of below equation based on the probability *Pi* probability and this is expressed as shown:

$$Pi = fiti / \sum_{n-1}^{SN} fit_n \tag{8}$$

In this equation, fit_i signifies the value of fitness solution y_i. The value of each generated candidate solution v_{ij} which is not in the boundary of search space allotted might be updated so that it will be within respective space.

The candidate solution fitness v_{ij} that are produced newly will be compared with the old ones. If the new solution has equal or better fitness values on comparing older ones in memory, this replaces preceding one's memory or else, the older ones are thus considered in memory. At the end of each search cycle, if the solution is not improved with the trial number that are predetermined, they are thus considered as a "limit" which will be exhausted; after that, the

solution is thereby abandoned through scout bees on searching new solutions in a random manner. The new solutions are thus generated from previous equation. By this, it was obvious that this scheme has three control parameters which are equal to the maximum cycles number (MCN) and the "limit" values.

For ensuring reproducibility and clarity, objective function employed in CBSO-aided feature optimization is defined explicitly. The optimization intends at selecting optimal feature subset which maximizes performance of classification on minimizing the redundancy among features. In specific, the objective function will be formulated as weighted combination of classification accuracy and compactness of feature subset.

$$F_{\text{obj}} = \alpha.Acc + (1-\alpha).\left(1 - \frac{|S|}{|F|}\right) \tag{9}$$

In this, Acc signifies classification accuracy attained with the use of selected feature subset S, |S| signifies the selected features number, |F| is the total number of features extracted, |F| denotes total number of extracted features, and $\alpha\epsilon[0,1]$ denotes the balancing parameter which controls trade-off among accuracy and feature reduction. This formulation ensures that this model prioritizes highly discriminative yet compact feature subsets, thus enhancing generalization of model and computational efficiency.

3.4 Classification using SAC-LSTM deep neural network approach

A self-attention coupled LSTM network comprises of one input layer, a hidden layer, and an output layer. The unit of common secret layer is regarded as the memory cubes that are not like existing NN (neural networks). Those blocks of memory consist of cells that saves time which were adapted several pairs of gating at the blocks for monitoring the flow of information. The input and output block activation are thus activated and is being regulated by 2 ways as input source and output source. Initially, the memory cell activation is done because of its presented SAC-LSTM which resolves the fading error issues thus maintain constant error. The selected feature is given as input as soon as the memory blocks are activated. The memory blocks are thus allowable for rearranging as the flow of information themselves turns out to be superfluous and an error input weight was, therefore, substituted by activation weight source. The sequence of training at which the backward and forward two separate networks are thus repeated and were linked to similar performance phase and it was considered as an undeveloped classifier principle presented. It suggests that entire grouping points before and after precise sequence of each point is having consecutive information that are extreme.

For enhancing the clarity and reproducibility, the model of SAC-LSTM has been briefly explained in revised manuscript. The suggested network comprises of stacked LSTM model with two hidden layers, every one consisting of 128 hidden units, enabling effectual temporal feature learning. An attention mechanism will be incorporated on LSTM layer top for emphasizing most relevant feature representations selectively. The attention module thus computes adaptive weights with the use of soft attention scheme, allowing the scheme in focusing on informative regions of feature sequence. At last, features attended are passed over fully connected layer followed by SoftMax classifier for final prediction. These architectural information offers clear understanding of design model and thus facilitate reproducibility.

The following are the steps in this classifier approach:

Algorithm 1 Pseudocode of SAC-LSTM

Step 1: Use the N training datapoints set that are random

Step 2: construct neural network depends on $N\ points$

Step 3: Select a precise duplicate from $phase\ 1\ and\ 2$ that are utilized for constructing classification.

Step 4: create feature that are correlated depending on a trusted value

This presented approach creates a group that are designated arbitrarily learning dataset zone of learning to make choices. It indicates that the classifier subsection thus votes as single work and becomes peaceful formerly on the unit of concluding test category. So as to manipulate the classified trusted values prediction features are to be computed. The proposed classifier SAC-LSTM parameters are thus intended as,

$$S_f(\text{i}) = \{ \begin{matrix} S_{f_{min}} + (S_{f_{max}} - S_{f_{min}}) * n, n < 1 \\ S_{f_{max}}/N_{iter}, \end{matrix}_{\text{i=1...n}} \tag{9}$$

In this, $f = fitness\ (i) - f_{min}$ which depends on present i[th] solution quality. The following analysis is done once the initialization of parameter is carried.

$$\sigma_v = (\gamma(1+\beta_S))*\sin(\pi*\beta_S/2)/(\gamma(1+\beta_S\ 2)*\beta_S{}^{\beta_c-1/2})*(\beta_S - 1/2)^{1/(\beta_S-1/2)} \tag{10}$$

In this, σ_v represents a random sword bee population.

Though LSTM networks are designed traditionally for sequential data, its application in this model is thus motivated by structured representation of image features as the ordered sequence. The features extracted will be arranged in a sequential manner, enabling LSTM for capturing long-range dependencies effectively and contextual relationships among feature elements which might not be exploited fully by traditional CNN models. Not like CNN, that focus

primarily on local spatial patterns, LSTM-aided models global inter-features correlations, improving discriminative ability. It is supported further by comparative analysis with CNN aided schemes, at which suggested SAC-LSTM shows enhanced performance in capturing complex feature interactions.

Thus, the classifier identifies and classifies breast cancer and its types as benign or malignant.

4 Performance analysis

This section offers a detailed presentation on the estimation of proposed approach performance. The analysis attained is related with traditional approaches and the estimated outcomes are projected here.

4.1 Dataset description

In this, BreakHis breast cancer dataset [21] is considered for input for estimating the proposed method robustness. This input dataset consists of about 7909 benign and malignant histopathological images that are from 82 patients. Histopathological images are thus acquired by dissimilar factors of magnification in this dataset such as $40x, 100x, 200x, and 400x$. For ensuring the validity of experimental design and prevent data leakage, the dataset was portioned with the use of patient level split strategy, at which images from similar patient were assigned exclusively to either testing or training sets. In addition, k-fold cross validation scheme was employed on training set for ensuring generalizability and robustness of model. This model prevents overlap among training and testing samples thereby ensuring that performance reported accurately reflects real-world generalizability ability.

A suggested model employs entire histopathological images that are available deprived of seeing magnification factors for classifying the sample classes images. Also, criteria of image level and patient level for sample dataset was employed. In an image-level approach, about 80% of independent magnification histopathological images are thus utilized for training and about 20% among them were utilized as test data (Fig. 2).

4.2 Performance and comparative estimation of proposed technique with optimal feature extraction method

Table 1 and Fig. 3 signify the comparison of proposed optimal feature extraction and classifier model with the existing feature extraction and classifier outcome. The comparison is made in terms of precision, recall, accuracy, F-measure, Gmean. The existing techniques employ SURF feature extraction model with various types of classifiers like KNN, Naïve Bayes, Discrete transform, SVM, GA-CNN, and proposed CBSO+SAC-LSTM. The outcomes attained reveal that the proposed model offers a higher rate of performance than others.

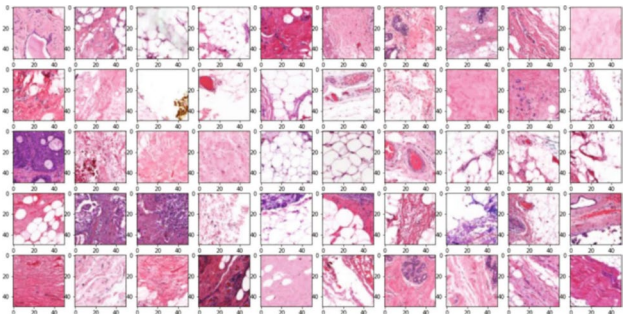


Fig. 2 Input sample image representation

Table 2 and Fig. 4 signify the comparison of proposed optimal feature extraction and classifier model with the existing feature extraction and classifier outcome. The comparison is made in terms of accuracy, precision. The existing techniques employ GLCM feature extraction model with various types of classifiers like KNN, Naïve Bayes, Discrete transform, SVM, GA-CNN, and proposed CBSO+SAC-LSTM. The outcomes attained reveal that the proposed model offers a higher rate of performance than others.

4.3 Performance and comparative analysis of proposed classifier method

The comparative evaluation made for proposed and traditional methods are projected in this section. The analysis is made to estimate the proposed method effectiveness.

A performance evaluation is carried in Table 3 and Fig. 5 for the proposed model and the outcomes are attained by accuracy, sensitivity, specificity, precision, F-score, and balanced accuracy (BAC) for the classifier designed. The attained outcomes are related with existing approaches and the analysis made reveals that a proposed model outcome is enhanced than traditional techniques to detect breast cancer.

A comparison is made for various existing techniques and proposed model and is shown in Table 4 and Fig. 6. The estimation is made to detect breast cancer with the use of similar histopathological image dataset provided in this work. A three-layer CNN method has an F-score rate of 71.8% having BAC of 84.23%. Likewise, for similar dataset Romero et al. suggested a DL method from inception manner. In this, F-score value of 89.7% and BAC of 89.00% was attained respectively. An AlexNet approach was presented by Janowczyk et al., and it has F-score value of 76.48% with BAC value 84.68%. Then, the author Reza et al. presented a patch dependent division of data which offers F-score of

Table 1 Comparison of proposed optimal feature extraction and classifier technique with existing extraction of feature and outcome of classifier [8]

Method	Accuracy	Precision	Recall	F-Measure	Gmean
SURF + KNN	66.12	63.12	71.44	63.75	61.91
SURF + NB	78.48	74.44	73.12	73.36	76.68
SURF + Discrete transform	82.11	83.52	81.13	82.72	82.11
SURF + SVM	86.12	85.83	86.78	81.37	82.75
SURF + GA-CNN	89.13	86.23	81.47	85.38	85.17
CSBO + SAC-LSTM	99.31	98.09	97.96	98.76	98.69

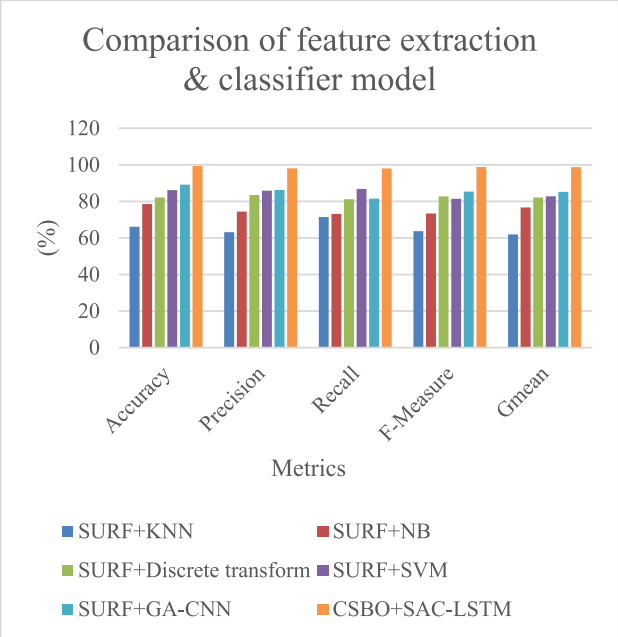


Fig. 3 Comparison of proposed optimal feature extraction and classifier technique with existing extraction of feature and outcome of classifier

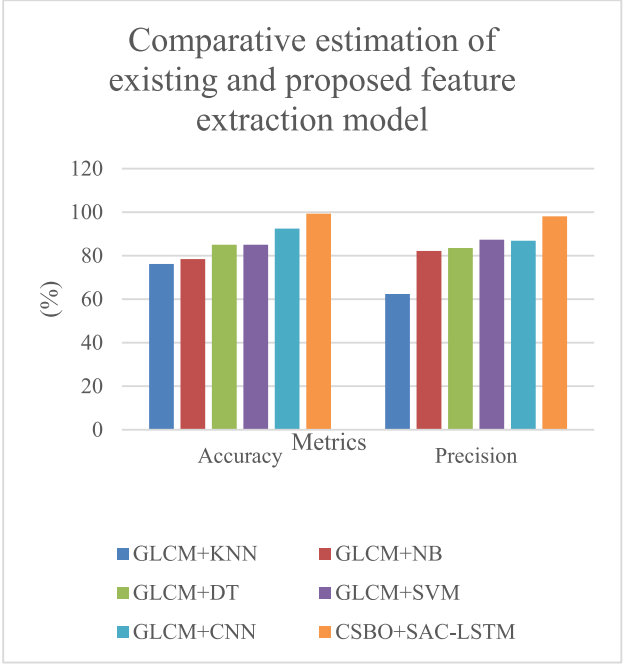


Fig. 4 Analysis of performance comparison of proposed and existing feature extraction and classifier model

Table 2 Analysis of performance comparison of proposed and existing feature extraction and classifier model [8]

Method	Accuracy	Precision
GLCM + KNN	76.17	62.4
GLCM + NB	78.45	82.16
GLCM + DT	85	83.56
GLCM + SVM	85	87.32
GLCM + CNN	92.44	86.89
CSBO + SAC-LSTM	99.31	98.09

84.78% with BAC performance of 85.48%. A traditional ResNet-50 technique provides 94.11% value of F-score, and 90.96% of BAC. Similarly, existing DenseNet-161 attains F-score value of 92.38% with BAC value as 91.57% BAC. The proposed classifier model SAC-LSTM 98.76% F-score

Table 3 Comparative evaluation for proposed and existing methods [22]

Metrics	Estimated performance		
	ResNet- 50	Densenet-161	Proposed (SAC-LSTM)
Accuracy	91.96	91.2	99.31
Specificity	88.28	93.56	97.96
Sensitivity	93.64	89.59	100
F-score	94.11	92.38	98.76
Precision	94.58	95.34	98.09
Balanced accuracy (BAC)	90.96	91.57	98.68

and 98.68% BAC value which is higher than the traditional models. Therefore, the proposed model is effective than other existing models.

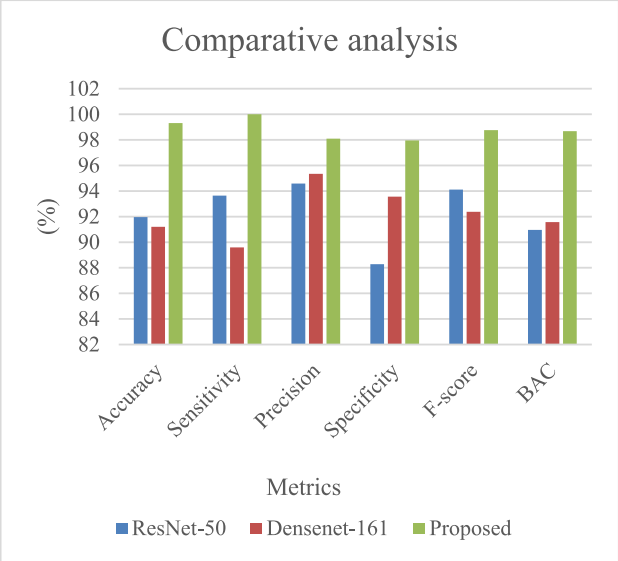


Fig. 5 Comparative evaluation of proposed and existing methods

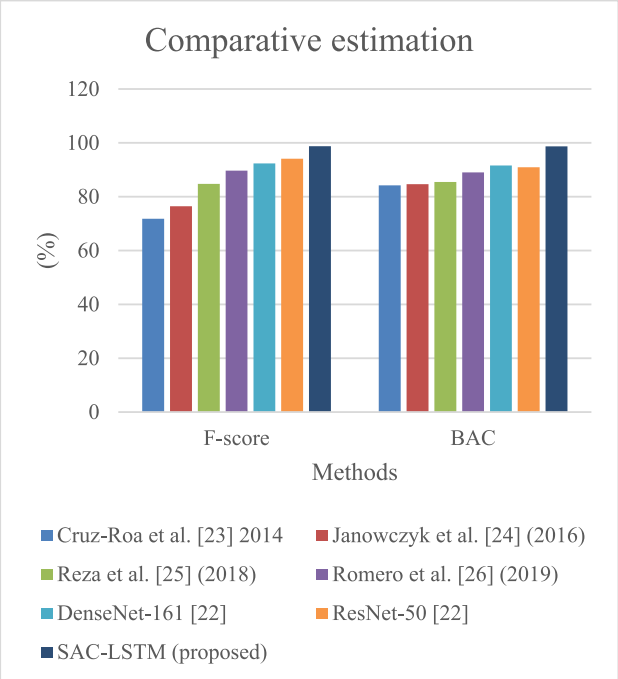


Fig. 6 Comparative estimation of proposed and existing models in terms of F-score and BAC

Figure 7 denotes the prediction map representation for the samples of histopathology image in testing data attained by means of SAC-LSTM classifier. The input original Histopathology images or whole slide image (WSI) with their annotated patches and its prediction maps are projected.

5 Conclusion

An improved SAC-LSTM with optimal CSBO-based feature extraction model was presented and evaluated which was effective in solving issues occurred at each step iteration step of the learning process at the sample selection. The optimization-based technique was employed to select aggregation of optimal texture, graph, and the morphological features. An enhanced classifier utilizes these combined features or characteristics as an input and functions well. The presented strategy was estimated, and the outcomes attained are compared with traditional methods with respect to accuracy, recall, precision, specificity, sensitivity, F-measure, BAC, and Gmean. The comparison of proposed feature extraction and classifier model integration was made, and outcomes are compared with traditional combinations of feature extraction and various classifier models. Likewise, the classifier model is compared with various existing classifier models. Then, the prediction map attained from input histopathology image by proposed SAC-LSTM classifier was projected. From these analyses, it was obvious that the proposed method performs well than other models and was said to be effective in offering higher detection rate with improved classification accuracy. The conclusion offers a clear summary of suggested objectives and outcomes. Future work might focus on extending the framework over external validation on multi-center datasets, thus exploring incorporation of

Table 4 Various existing models compared with proposed techniques in terms of F-score and BAC

Reference	Size of Input	Technique	F-score	BAC
Cruz-Roa et al. [23]	50 × 50	3-layer CNN	71.8	84.23
Janowczyk et al. [24]	32 × 32	AlexNet	76.48	84.68
Reza et al. [25]	50 × 50	SMOTE + CNN	84.78	85.48
Romero et al. [26]	50 × 50	Multi-level batch normalization	89.7	89
DenseNet-161 [22]	50 × 50	DenseNet-161 transfer learning	92.38	91.57
ResNet-50 [22]	50 × 50	ResNet-50 transfer learning	94.11	90.96
SAC-LSTM (proposed)	50 × 50	Proposed SAC-LSTM	98.76	98.68

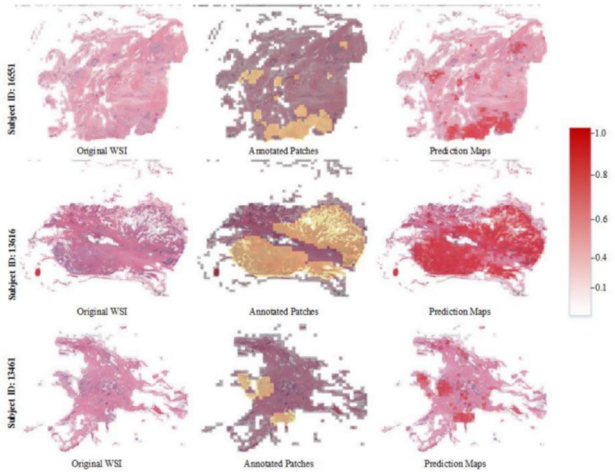


Fig. 7 Input test data samples of histopathology images with prediction maps attained by SAC-LSTM approach

clinical decision support systems and, thus, improving its practical impact and applicability.

Author contribution Author contribution- All authors contributed to the study's conception, design, material preparation, data collection, and analysis. All authors read and approved the final manuscript.

Funding Not Applicable.

Data availability No datasets were generated or analyzed during the current study.

Declarations

Competing interest The authors declare no competing interests.

Research involving human and /or animals Not Applicable.

Informed consent Not Applicable.

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